Emotion Detection and Music Recommendation

AI-Based Emotion Recognition for Personalized Music Experience

Moodify
2026| AI & ML Project
Infosys Springboard

Emotion Detection for Music

What is Emotion Detection

Emotion detection is the process of identifying
Human emotions from facial expressions using AI
And Computer Vision .It analyzes micro-expressions,
Facial landmarks , and temporal cues to infer
Emotional states in real-time.

Why Emotions Matter in Music

Emotional states directly influence music preferences;
Real-time emotion detection enables adaptive playlists and
Personalized recommendations, enhancing user engagement and
Creating more emotionally resonant listening experiences.

Problem Statement

Time-Consuming Process

Manual playlist selection demands repeated searching and setup, consuming users' valuable time.

Frequent Mood Changes

Users' emotions shift throughout the day, making static playlists often inappropriate quickly.

No Real-Time Personalization

Current systems lack adaptive adjustments based on immediate context and user feedback.

Proposed Solution

1. **User Input**

   Upload an image or capture live from webcam for analysis.

2. **Emotion Detection**

   AI analyzes facial expressions to infer users' current emotions.

3. **Music Recommendation**

   System suggests songs matching detected mood and user preferences.

4. **Spotify Integration**

   Optional redirect to Spotify for seamless playback and saving.

System Features

Real-Time Emotion Detection

Performs instant facial expression analysis to identify emotions in real time for adaptive responses accurately

Image Upload & Webcam

Supports uploading images or using webcam stream, enabling flexible dual input methods for emotion capture

Emotion-Based Music Mapping

Maps detected emotions to suitable genres, intelligently recommending personalized music that matches users' mood preferences

Web Interface

Offers a responsive, user-friendly web interface for easy emotion analysis, settings, and result visualization quickly

FER-2013 Dataset Overview

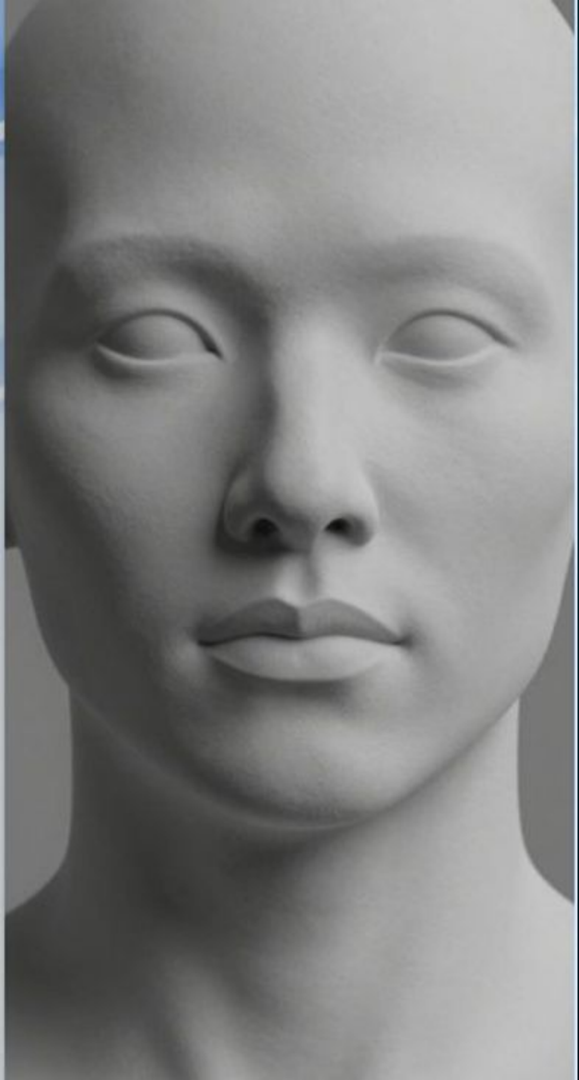
FER-2013(Facial Emotion Recognition 2013) is a benchmark dataset containing 35,887 grayscale facial images, each 48x48 pixels.The dataset covers 7 emotion classes: Angry, Disgust, Fear, Happy, Neutral, Sad and Surprised. It is widely used for training and evaluating emotion recognition models in computer vision and deep learning research.

Dataset Characteristics

Contains 35,887 grayscale 48x48 images across 7 labeled emotion classes, designed for classification tasks and model benchmarking.

Dataset Relevance

Widely adopted in academia and industry for benchmarking facial expression classifiers and advancing deep learning-based emotion recognition research.



Data Preprocessing Pipeline

Below are the Sequential preprocessing steps applied to the dataset:

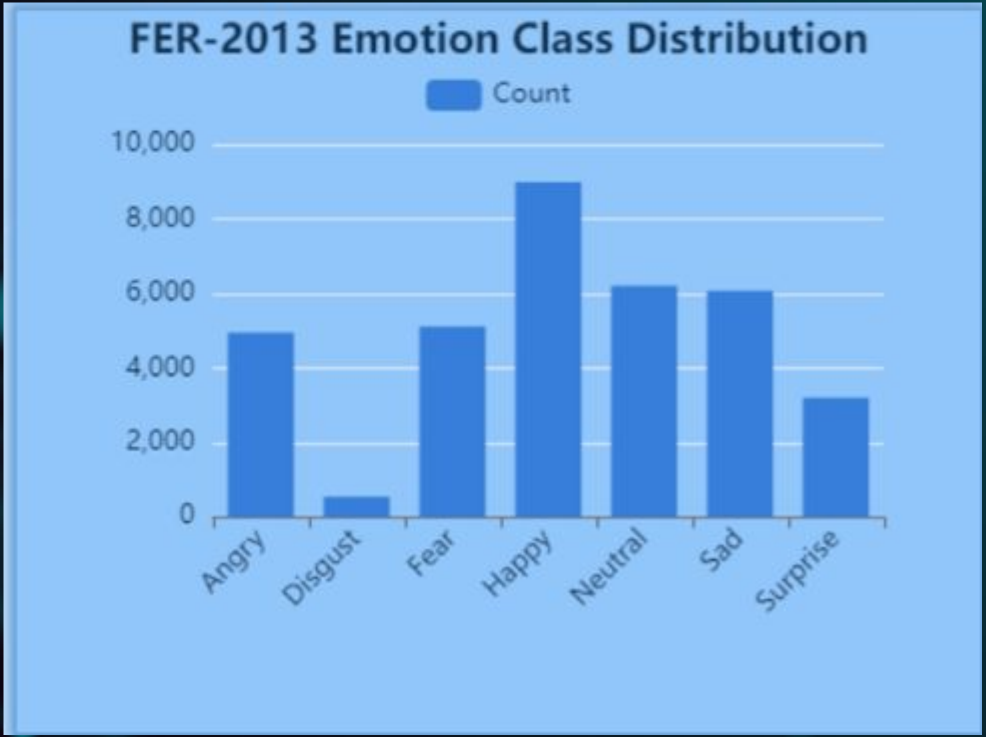
1.GrayScale Normalization - Converting images to standard pixel intensity values(0-255) for consistency across the dataset, ensuring uniform input scale for the model.

2.Face Resizing- Standardizing all images to 48x48 pixels as required by the model , preserving aspect ratio and centering the face region where possible.

3. Data Augmentation and Noise Removal - Applying rotation , zoom, and noise filtering techniques such as Gaussian smoothing and median filtering to enhance dataset robustness and reduce artifacts.

Right : after preprocessing – normalized and cleaned facial image suitable for model input.

Emotion Distribution Analysis



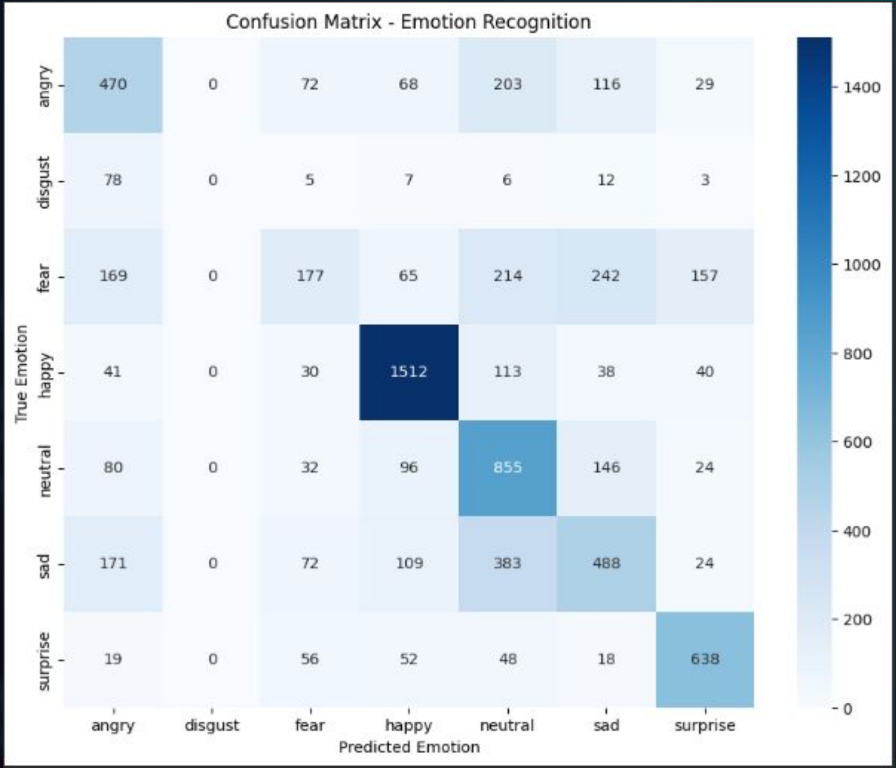
Dataset Balance

Happy emotion dominates with 8,989 images. Disgust significantly underrepresented at 547 images, requiring careful handling during training.

Class Imbalance Impact

Imbalanced data can bias model predictions. Weighted loss functions and oversampling techniques are applied to address this challenge.

Confusion Matrix



The confusion matrix compares **actual emotions** with **predicted emotions**.

Rows represent true emotion labels.

Columns represent model predictions.

Diagonal values show correct classifications.

Off-diagonal values indicate misclassifications.

Happy emotion shows the highest accuracy with **1512 correct predictions**.

Surprise and **Neutral** emotions are also classified accurately.

Fear and Sad emotions show confusion with **Neutral and Angry**, due to similar facial features.

Disgust has very few samples, leading to lower performance.

Overall, the model performs well for **distinct emotions** and struggles with **subtle expressions**.

CNN Model Architecture

A concise, vertical step-based overview describing data flow through the network.

Input Layer

48×48 grayscale images normalized and reshaped into tensors for model input.

Convolutional Blocks

Four to five convolutional layers extract spatial features using ReLU activation.

Max Pooling Layers

Interleaved max pooling reduces spatial dimensions and increases receptive field.

Fully Connected Layers

Dense layers with 256 units and dropout applied for regularization and robustness.

Output Layer

Seven neurons with Softmax activation to produce emotion class probabilities.

System Workflow

1

Step 1: User Input

User uploads image file or grants webcam access.

2

Step 2: Face Detection

Algorithm locates face and extracts precise facial region.

3

Step 3: Emotion Prediction

CNN processes the face image and outputs emotion class.

4

Step 4: Music Recommendation

System maps predicted emotion to appropriate music genre.

5

Step 5: User Output

Display recommended playlist with direct Spotify link provided.

Music Recommendation Logic

We translate detected user emotions into musical attributes by mapping affective states to genres, tempos and moods to drive personalized recommendations and listening experiences.

Emotion Classification

The model outputs seven emotions: happy, sad, angry, relaxed, surprised, fearful, and neutral, each illustrated with brief examples.

Genre Mapping

Each detected emotion is mapped to preferred genres; for example, Happy to Pop and Dance, Sad to Blues and Ballads.

Playlist Integration

Curated static playlists provide handpicked tracks per emotion; optionally integrate Spotify API for dynamic, personalized recommendations in real time.

Results and Output

94 %

Sample Prediction

Confidence Score 94% – Detected emotion:Happy, facial crop analyzed with detected smile landmarks and high confidence in positive affect.

56 %
Training
Accuracy

Model Training Accuracy : 56% measured on training dataset after final epoch.

54 %
Validation
Accuracy

Model Validation Accuracy : 54% measured on hold-out validation set, indicating room for generalization improvement.

Challenges & Limitations

Imbalanced Dataset

Some emotions (e.g., *Disgust*) have very few samples, affecting learning accuracy.

Similar Facial Expressions

Emotions like *Fear, Sad, and Neutral* have overlapping features, causing misclassification.

Lighting & Image Quality Issues

Variations in lighting and low-resolution images reduce detection performance.

Real-Time Processing Constraints

Maintaining high accuracy during live webcam emotion detection is challenging.

Limited Generalization

Model performance may drop for faces not well represented in the training data

Limited Music Dataset
A small, manually curated dataset restricts recommendation variety.

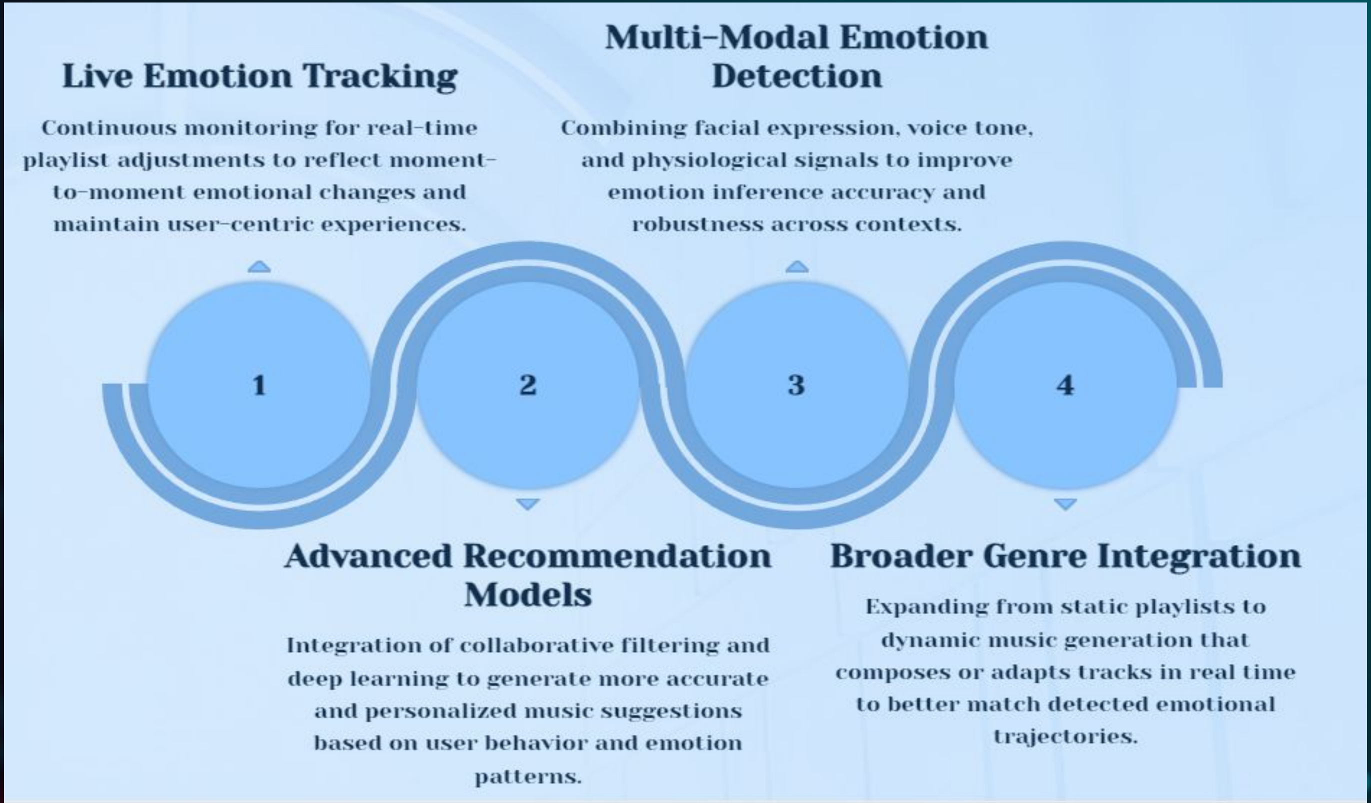
Static Emotion–Music Mapping
Fixed emotion-to-song mapping may not suit all users.

Lack of Personalization
User preferences, listening history, and feedback are not considered.

Subjective Nature of Music Emotion
The same song can evoke different emotions for different users.

No Real-Time Adaptation
Music recommendations do not change dynamically with emotion shifts.

Future Scope



Conclusion

- The system successfully bridges emotion recognition and personalized music recommendation by detecting user's emotional states and mapping them to tailored playlists. Through real-time analysis and contextual understanding . It enhances user engagement and delivers more emotionally resonant listening experiences.

THANK YOU!!!